Lightweight Spike-Based DOA Sector Classification for Embedded Edge Inference

Hossein Molhem
Department of Electrical and Computer Engineering
Kennesaw State University
Email: hmolhem@students.kennesaw.edu

***Abstract*—This paper presents a compact Python reference prototype for direction-of-arrival (DOA) sector classification using a physics-based uniform linear array (ULA) signal model and a leaky-integrate-and-fire (LIF)-inspired inference path. The objective is not to introduce a new DOA estimator, but to establish a reproducible embedded-AI baseline that connects array signal processing, sparse spike encoding, and latency-aware temporal inference. Beamspace energy features are extracted from sample covariance matrices and classified using both a nearest-centroid non-spiking baseline and a rate-coded LIF-inspired classifier. The nearest-centroid baseline achieves an overall accuracy of 86.90%. A fixed LIF configuration achieves 82.50% accuracy, while a time-window/leak parameter sweep improves the best LIF accuracy to 84.17%. The spike-based path is not presented as an accuracy-improving replacement for the continuous baseline. Its added value is architectural: it converts continuous beamspace inference into a sparse temporal decision process with measurable stable-decision latency and tunable accuracy–latency behavior, which is suitable for later embedded implementation studies.**

***Index Terms*—Embedded AI, spiking neural networks, LIF neuron, DOA classification, uniform linear array, edge inference, beamspace features.**

I. Introduction

Edge sensing systems often require inference under constrained compute, memory, and latency budgets. For such systems, lightweight and hardware-aware inference methods can be preferable to large black-box neural networks. Spike-based inference is attractive because information can be represented through sparse temporal events, potentially reducing arithmetic activity and supporting early decision logic [4], [6].

This work uses DOA sector classification as a controlled sensing problem. The selected problem is technically relevant because it is grounded in array signal processing, yet compact enough to support an end-to-end Phase 1 prototype. The contribution is an engineering baseline that compares continuous-valued beamspace classification with a sparse LIF-inspired temporal inference path.

II. Signal Model

A narrowband single-source ULA model is used. For source angle θ, the received snapshot matrix is

$$X = a(\theta)s + N, \tag{1}$$

where $a(\theta)$ is the steering vector, s is the source signal, and N is circular complex Gaussian noise. The sample covariance matrix is

$$R_{xx} = \frac{1}{M} XX^H, \tag{2}$$

where $M = 128$ snapshots.

Beamspace energy features are extracted over a fixed angular grid. The classification target is not the exact continuous angle, but one of four coarse angular sectors spanning -60° to $+60^\circ$.

III. Theoretical Foundation

For a ULA with element spacing d and wavelength λ, the steering vector under the broadside convention can be written as [1]

$$a(\theta) = [1, e^{-jkd\sin\theta}, \ldots, e^{-j(N-1)kd\sin\theta}]^T, \tag{3}$$

where $k = 2\pi/\lambda$. This phase progression is the physical mechanism that makes the covariance matrix angle-dependent.

For a steering vector $b(\phi)$ evaluated at grid angle ϕ, the beamspace energy feature is represented as

$$p(\phi) = b^H(\phi) R_{xx} b(\phi). \tag{4}$$

Evaluating $p(\phi)$ over a fixed angular grid converts the complex covariance matrix into a compact real-valued feature vector suitable for sector classification [1], [2].

The non-spiking baseline uses the nearest-centroid decision rule

$$\hat{c} = \arg\min_c \|x - \mu_c\|_2^2, \tag{5}$$

where x is the normalized beamspace feature vector and μ_c is the centroid of class c. This rule is intentionally simple and interpretable, making it useful as a reference classifier [3].

For rate-based spike encoding, each normalized feature value $z_i \in [0, 1]$ is interpreted as the probability of spike emission at each time step. The LIF-inspired classifier then accumulates class-level evidence as

$$v_c[t] = \alpha v_c[t-1] + i_c[t], \tag{6}$$

where α is the leak factor and $i_c[t]$ is the spike-driven input current. This model is used as a hardware-facing abstraction of temporal evidence accumulation rather than as a biologically detailed neuron model [5], [6].

IV. Methodology

A. Dataset Configuration

The dataset contains 100 samples per sector per SNR value. A 70/30 train/test split is used with stratification by SNR and class.

TABLE I
DATASET CONFIGURATION

Parameter	Value
Array elements	8
Snapshots	128
Spacing	$\lambda/2$
DOA range	-60° to $+60^\circ$
Sectors	4
SNR values	-10 to 20 dB
Total samples	2800
Test samples	840

TABLE II
KEY RUN 005 SWEEP RESULTS

Case	T	Leak	Acc.	Lat.
Best LIF accuracy	40	0.99	0.841667	8.396429
Lowest latency	10	0.99	0.776190	3.782143
Fixed reference	20	0.97	0.825000	6.423810
Baseline	–	–	0.869048	–

B. Nearest-Centroid Baseline

The baseline classifier is a nearest-centroid classifier applied to normalized beamspace feature vectors. This classifier is deterministic and interpretable, making it an appropriate reference point for measuring the effect of spike encoding.

C. Rate-Based Spike Encoding

Normalized features are converted into binary spike trains using rate-based encoding. Higher normalized feature values correspond to higher spike probabilities over a finite time window.

D. LIF-Inspired Classifier

The LIF-inspired classifier is implemented as a leaky membrane accumulator for each class:

$$v_c[t] = \alpha v_c[t-1] + i_c[t], \tag{7}$$

where $v_c[t]$ is the class membrane state, α is the leak factor, and $i_c[t]$ is the spike-driven input current. The class with the largest final membrane value is selected. Stable-decision latency is estimated as the earliest time after which the final decision remains unchanged.

V. RESULTS

The nearest-centroid baseline achieved an overall accuracy of 0.869048. The fixed LIF configuration with $T = 20$ and leak $= 0.97$ achieved 0.825000 accuracy and a mean stable-decision latency of 6.423810 time steps.

Run 005 swept the number of time steps and membrane leak factor. Table II summarizes the most important cases.

VI. DISCUSSION

The results show that compact beamspace features are informative for sector-level DOA classification. The non-spiking nearest-centroid classifier achieves the highest accuracy, as expected, because it operates directly on continuous-valued normalized features. The LIF-inspired classifier incurs some accuracy loss because the features are converted into sparse stochastic spike trains. However, this loss is moderate, and the LIF path provides a tunable temporal decision process.

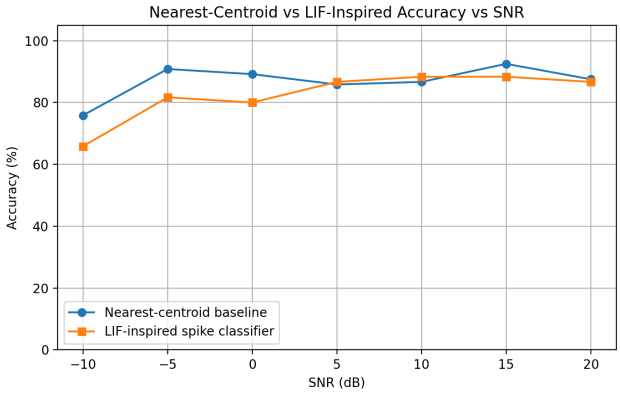


Fig. 1. Accuracy comparison between the nearest-centroid baseline and the fixed LIF-inspired classifier.

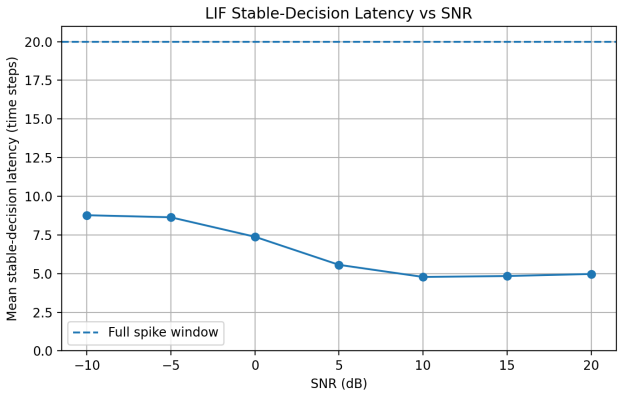


Fig. 2. Mean stable-decision latency versus SNR for the LIF-inspired classifier.

The parameter sweep confirms that longer time windows generally improve accuracy, while shorter time windows reduce mean stable-decision latency. Higher leak values preserve evidence more effectively in this prototype. This behavior is useful for embedded inference because it exposes a design knob between accuracy and latency.

VII. CONCLUSION

A complete Phase 1 Python prototype was developed for DOA sector classification using a physics-based ULA model, beamspace features, nearest-centroid baseline classification, rate-coded spike trains, and LIF-inspired temporal inference. The spike-based path did not improve raw accuracy over the continuous nearest-centroid baseline: the baseline achieved 86.90%, the fixed LIF configuration achieved 82.50%, and the best LIF sweep configuration reached 84.17%.

The value of the spike-based formulation is therefore architectural rather than purely accuracy-based. It transforms continuous beamspace inference into a sparse temporal decision

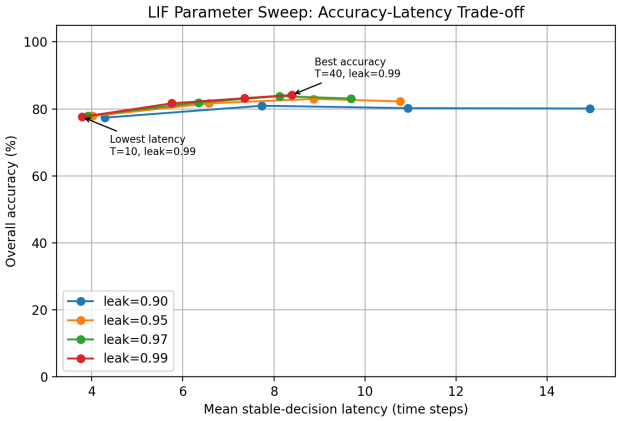


Fig. 3. Accuracy–latency trade-off for the LIF time-window and leak sweep.

process, exposes stable-decision latency as a measurable output, and provides tunable accuracy–latency behavior through the spike time window and leak factor. The result establishes a reproducible hardware-aware embedded-AI reference path for later work involving richer radar/DOA cases, PYNQ deployment, and FPGA-oriented acceleration.

REFERENCES

[1] H. L. Van Trees, *Optimum Array Processing*. Wiley, 2002.
[2] S. Haykin, *Adaptive Filter Theory*. Pearson, 2014.
[3] R. O. Duda, P. E. Hart, and D. G. Stork, *Pattern Classification*. Wiley, 2001.
[4] W. Maass, "Networks of spiking neurons: The third generation of neural network models," *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997.
[5] W. Gerstner and W. M. Kistler, *Spiking Neuron Models: Single Neurons, Populations, Plasticity*. Cambridge University Press, 2002.
[6] K. Roy, A. Jaiswal, and P. Panda, "Towards spike-based machine intelligence with neuromorphic computing," *Nature*, vol. 575, pp. 607–617, 2019.